



Passenger-Centred Evaluation of Digital Railway Transport Systems: The Mediating Roles of Perceived Value and Digital Trust



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Abstract: Digital railway transport systems increasingly integrate online ticketing, real-time passenger information, digital payment, onboard connectivity, and customer support into the passenger journey. Their effectiveness depends not only on the availability of these functions but also on how passengers experience and evaluate their interactions with them. This study investigates the relationship between digital experience and railway passenger satisfaction (RPS) and examines the parallel mediating roles of perceived value and digital trust in the Vietnamese railway context. Survey data were collected from 348 passengers who had recently used at least one digital railway service and were analysed using partial least squares structural equation modelling. Digital experience was positively associated with passenger satisfaction ($\beta = 0.452, p < 0.001$) and accounted for significant variation in perceived value and digital trust. Both mediating pathways were statistically significant, although the indirect effect through perceived value ($\beta = 0.175, p < 0.001$) was stronger than that through digital trust ($\beta = 0.113, p = 0.001$). Together, digital experience, perceived value, and digital trust explained 62.2% of the variance in RPS. These findings indicate that the passenger-side performance of digital railway transport systems rests on the quality of integrated interactions across digital touchpoints, particularly their convenience, reliability, security, and service value. The study provides a passenger-centred framework for evaluating the implementation of digital railway services and identifies the user-related factors that railway operators should consider when planning and prioritising digital system improvements.

Keywords: Digital railway transport systems; Passenger-centred evaluation; Digital experience; Railway passenger satisfaction; Perceived value; Digital trust

1 Introduction

Rail transport is widely recognized as an efficient, climate-friendly mode for both passenger and freight movement. Nevertheless, railways have experienced a steady decline in their share of the overall transport market in many countries, including Vietnam [1]. Digitalization offers one route by which railway operators can respond to this challenge. Contemporary railway transport systems increasingly incorporate online ticketing, electronic payment, real-time passenger information, onboard connectivity, and digitally mediated customer support. These functions form the user-facing layer of the transport system and connect passengers with information, transactions, and operational services before, during, and after a journey. Their contribution cannot be judged solely by whether the relevant technologies have been installed. It also depends on whether passengers can use them conveniently, rely on the information and transactions they provide, and obtain value from the resulting interaction. A passenger-centered evaluation is therefore necessary to determine how the implementation of digital railway systems is reflected in perceived value, trust, and satisfaction. These user-side outcomes matter to the quality and competitiveness of railway transport because they indicate whether digital functions operate as a coherent part of the passenger journey rather than as a collection of isolated tools.

Since 2020, the Vietnam Railways Corporation (VNR) has introduced mobile online ticketing and payment systems that allow passengers to select trains, reserve tickets, and choose payment methods through a digital interface. The service environment now includes digital wallets and specialized applications such as Momo, the "Ve Tau" App, VNPAY, Vimo e-wallet, and Viettel Money. VNR has also integrated its passenger services with the

Zalo ecosystem. Taken together, these platforms represent an expanding digital interface between passengers and the railway system, intended to make travel arrangements safer, more convenient, and more comfortable.

Within this ecosystem, the Zalo Notification Service (ZNS) uses passengers' telephone numbers to deliver booking confirmations and real-time itinerary information through the Zalo messaging interface. The Zalo Official Account (Zalo OA) provides an authorized channel through which VNR can communicate with passengers on Vietnam's most widely used messaging platform. It serves as an information point for train schedules and fares and allows passengers to obtain departure information and relevant travel rules. The embedded Zalo Mini App also permits passengers to book journeys and receive electronic tickets without leaving the application. During service disruptions, the same channels can communicate the status of an incident, remedial action, and instructions for alternative travel arrangements. The ecosystem therefore combines transaction, information, and disruption-response functions within a single passenger-facing environment.

VNR has concurrently incorporated other applications into its service delivery framework. Its multi-platform payment environment combines different payment options with promotional incentives and is intended to improve transaction convenience and security. Passenger information is distributed through Light Emitting Diode (LED) displays at stations, while operational updates are issued through Short Message Service (SMS) and email. Onboard Wi-Fi and entertainment services extend the digital environment into the train itself. Customer care has likewise been digitized through social media and messaging platforms that receive passenger feedback and provide individual support. These facilities span the principal stages of a journey, but their technical availability does not in itself show whether passengers experience the combined system as useful, dependable, and responsive.

Existing railway research commonly examines individual digital functions rather than passengers' integrated experience across the full journey. As a result, there is limited evidence on how passengers evaluate a railway system in which ticketing, payment, information, connectivity, and support are delivered through interconnected digital touchpoints. Perceived value and digital trust are also often analyzed separately. This makes it difficult to determine whether satisfaction arises primarily because passengers consider the system worthwhile or because they consider it reliable and secure. The relative importance of these two mechanisms remains insufficiently understood. Evidence from Vietnamese railway passenger services is particularly limited despite the rapid expansion of the country's digital platforms. This study addresses these gaps by treating Digital Experience as a unified representation of passengers' interactions with the user-facing components of the digital railway transport system. Using survey data from railway passengers, it examines perceived value and digital trust as parallel mediators between digital experience and railway passenger satisfaction (RPS). The purpose is to provide a passenger-centered evaluation of digital system implementation and evidence that can guide the prioritization of digital improvements in an evolving railway market.

Four complementary theoretical perspectives explain the relationship between digital experience and passenger satisfaction and the mediating roles of perceived value and digital trust. Together, they connect passengers' interaction with a digital railway system to their evaluation of its usefulness, quality, value, and dependability.

The technology acceptance model (TAM) [2] explains the acceptance and use of technology through perceived usefulness and perceived ease of use. Rooted in the Theory of Reasoned Action and information systems research, TAM proposes that the decision to use a system depends on whether users consider it useful and on the effort required to operate it. Perceived usefulness refers to the expectation that a system will improve task performance, whereas perceived ease of use concerns the effort involved in using it. These assessments shape attitudes toward a technology and, subsequently, actual use behavior [3, 4]. TAM has been applied across a wide range of domains and geographical settings and remains an established framework for explaining user behavior [5]. In railway transport, it helps explain how passengers assess their interaction with digital touchpoints throughout a journey. Its adaptable structure [6] also permits the inclusion of factors that are important in public transport, including information quality, system reliability, perceived risk, and digital trust. Such factors are especially relevant in railway operations, where failures in information or transactions may affect travel decisions beyond the digital interface itself. In this study, TAM therefore provides the basis for explaining how passengers' experience of digital railway functions shapes their subsequent evaluations.

The theory of customer perceived value [7] provides a second perspective on post-use evaluation. Perceived value is a passenger's overall judgment of service utility based on what is received in relation to what is given. This judgment weighs financial cost against non-monetary sacrifices, including time and mental effort [8]. The Perceived Value (PERVAL) scale [9] develops this reasoning through emotional, functional, social, and economic dimensions. In railway transport, digital interactions can create value by saving time, reducing uncertainty, simplifying transactions, and giving passengers greater control over their journeys. They may also impose costs when an interface is difficult to use, information is inconsistent, or a transaction fails. Perceived value therefore captures the passenger's overall assessment of the returns obtained from interacting with digital railway services. When these returns outweigh the monetary and non-monetary costs, passengers are more likely to be satisfied and to continue using the service. Perceived value consequently provides a mediating link between the quality of digital experience and passenger

outcomes.

Trust is also central to interaction in a digital environment. Users must be able to rely both on the service provider as an organization and on the operational capability of the technological system [10]. Organizational policies, commitments, and protection mechanisms can support digital trust, but passengers encounter those commitments through the actual performance of the system. Accurate transactions, consistent information, secure payment, and appropriate handling of personal data make the system easier to rely on. Trust also reduces perceived risk when users face the uncertainty associated with online interfaces [11]. Ladhari [12] shows that trust is related not only to satisfaction following a positive service experience but also to the relationship between perceived service quality and long-term usage intentions. Digital trust may therefore transmit part of the relationship between passengers' digital experience and RPS by reducing uncertainty and supporting a favorable evaluation of system interaction.

Service quality theory, represented by the Service Quality (SERVQUAL) model [13], defines service quality through the gap between customers' expectations and their perceptions after a service encounter. SERVQUAL distinguishes five dimensions: tangibles, reliability, responsiveness, assurance, and empathy. These dimensions describe how users judge a provider's performance and the service encounter as a whole. In a digital setting, this framework has been extended to include system, information, and service quality. The extension accounts for system integrity, the reliability and accuracy of information, and the support available through digital channels [14, 15]. This distinction is important for railways because passengers experience digital functions as part of the wider transport service, not as an independent information system. Railway service quality must therefore be assessed through a combination of conventional service performance and the quality of the digital experience [16].

These perspectives address different stages of the same passenger–system relationship. TAM explains the initial assessment of usefulness and ease of use. Extended service quality theory identifies the system reliability, information accuracy, responsiveness, assurance, and support through which digital performance is experienced. Perceived value explains whether the resulting advantages justify the monetary and non-monetary costs, while digital trust explains whether passengers are willing to depend on the provider and its technological system under uncertainty. Their integration connects the performance of user-facing railway technologies with two distinct post-use evaluations—whether the digital service is worthwhile and whether it is dependable—through which digital experience is associated with RPS.

Empirical studies show that digital platforms in public transport are associated with passenger experience and satisfaction [17]. In railway transport, the digital experience extends across online ticketing and reservations, real-time passenger information, smart payment and ticket validation, onboard connectivity, and digital customer support. These functions correspond to different stages of the journey but are experienced by passengers as parts of the same transport system. Online booking and reservation applications can reduce physical queuing and make travel arrangements more flexible, convenient, and transparent [18]. Contactless smart ticketing can likewise simplify purchasing and improve the efficiency of the service encounter [19, 20].

Passenger information systems provide a second point of interaction. Timely and accurate updates on train schedules, platform changes, and expected arrival and departure times can support trust and improve the travel experience [21]. The delivery of real-time information through mobile applications has also been associated with higher passenger satisfaction [22]. By contrast, unreliable information can produce a sharp decline in satisfaction, particularly among passengers prone to travel anxiety [23, 24]. Access to digital cues and information throughout a journey can also provide greater psychological stability and reduce stress [25, 26]. Information quality is thus not merely an interface feature; it affects how passengers interpret and respond to the operation of the transport system.

The onboard digital experience is especially important in rail transport because trains are widely used for long-distance travel [27]. Seamless internet access through onboard Wi-Fi and related connectivity has been shown to influence the satisfaction of long-distance passengers [28, 29]. During an extended journey, passengers may use their travel time for work, entertainment, and social communication. Stable connectivity is therefore increasingly treated as part of the onboard service rather than as an optional amenity. Complimentary Wi-Fi and infrastructure that supports passengers' own devices can make railway services more attractive and may influence modal choice [30]. Availability and stability are central to the quality of this function. Calastri et al. [31] found that reliable connectivity encouraged participation in online activities and more productive use of travel time, which in turn contributed to the value assigned to the journey and to the overall evaluation of the travel experience.

Information technology also allows customer care communications to be adapted to individual needs and preferences, shaping passengers' assessment of experience quality [20]. Artificial intelligence (AI) and Big Data can support personalized itinerary planning and proactive management of service disruptions. These applications give passengers greater access to relevant information and more control over their journeys, with consequences for satisfaction [32, 33]. Across these settings, digital touchpoints connect passengers to the transaction, information, connectivity, and support functions of the railway system. When those interactions reduce effort and uncertainty and allow passengers to manage their journeys more effectively, the digital experience is expected to be positively related to their overall assessment of the railway service. The first hypothesis is therefore stated as follows:

H1. Digital Experience has a direct and positive effect on RPS.

Digital experience may be related to passenger satisfaction not only directly but also through perceived value and digital trust. Perceived value captures the passenger's assessment of the balance between the advantages received—including convenience, time savings, reliability, and comfort—and the associated costs, such as fares, waiting time, effort, perceived risk, and other inconveniences. Digital trust concerns the passenger's willingness to rely on the digital touchpoints and the organization responsible for them. A well-integrated digital experience can increase perceived value through convenience, transparency, and lower time and effort, while supporting trust through system stability, secure transactions, punctuality, and consistent information. Perceived value is one mechanism through which technological utility is translated into passenger satisfaction and loyalty [34, 35]. Passengers who find that the advantages of digital railway services outweigh their monetary and non-monetary costs are more likely to report higher satisfaction. This reasoning leads to the following hypotheses:

H2a. Digital experience has a positive effect on perceived value.

H2b. Perceived value has a positive effect on RPS.

H2. Perceived value mediates the relationship between digital experience and RPS.

Evidence from Thailand reported by Wonglakorn et al. [36] identifies perceived value and digital trust as important elements in the attitudinal and behavioral evaluation of railway services. Digital trust has also been positively related to RPS in interactions conducted through digital platforms [37]. Passengers who trust the digital environment are likely to regard such interactions as less risky and less psychologically demanding, which may lead to a more favorable assessment of the service as a whole. Research on the Doha Metro [38] similarly identifies trust as an important factor in experience evaluation and in the acceptance of technology-intensive urban rail services. These findings suggest that digital trust reduces the uncertainty associated with the use of passenger-facing technologies and thereby links digital experience to satisfaction. The corresponding hypotheses are as follows:

H3a. Digital experience has a positive effect on digital trust.

H3b. Digital trust has a positive effect on RPS.

H3. Digital trust mediates the relationship between digital experience and RPS.

Perceived value and digital trust are examined as parallel mediators because they represent distinct but complementary assessments of the same passenger–system interaction. Perceived value concerns whether digital railway services produce worthwhile returns, whereas digital trust concerns whether the services and the system supporting them are reliable and secure. Comparing the two pathways makes it possible to determine which aspect of passenger evaluation plays the larger role in connecting digital experience with RPS. The resulting conceptual model is presented in Figure 1.

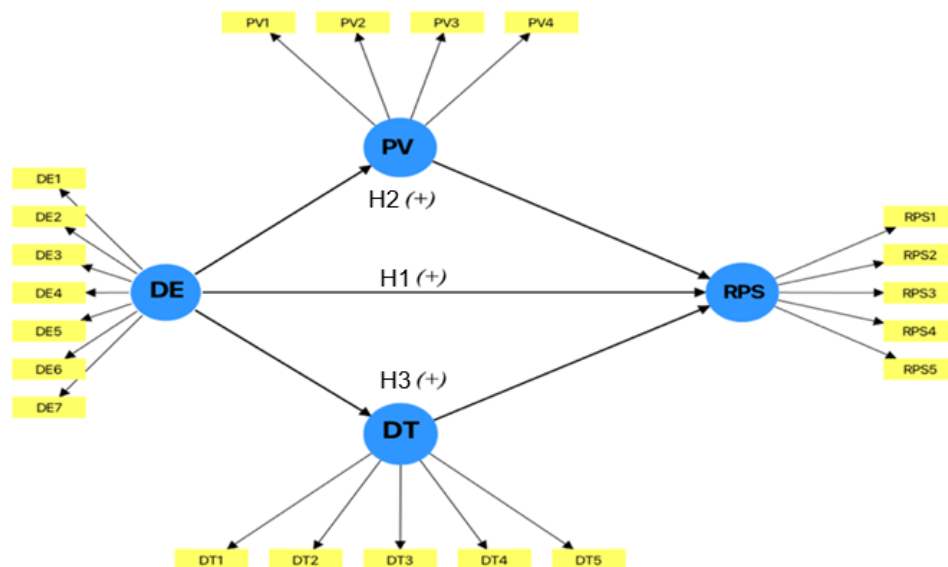


Figure 1. Model and hypotheses

Note: DE—digital experience; PV—perceived value; RPS—railway passenger satisfaction; DT—digital trust.

2 Methodology

A quantitative research design and a structured questionnaire were used to examine the relationship between passengers' digital experience and railway service satisfaction, with perceived value and digital trust specified as parallel mediators. The target population consisted of passengers who had used at least one digital service provided

by VNR. Eligible participants included both frequent and occasional passengers who had traveled on the Vietnamese railway network during the preceding six months. Non-probability purposive sampling was employed to obtain responses from participants with relevant experience [39]. Snowball sampling was subsequently used to extend recruitment to other eligible passengers. Initial respondents were asked to identify or invite individuals who met the participation criteria, and this process continued until the target sample size was reached [40, 41]. No age-based quotas or stratified sampling procedures were applied because eligibility was determined by recent use of at least one digital railway service. The age distribution therefore reflected the passengers reached through purposive and snowball recruitment rather than a predetermined allocation across age groups. This procedure ensured that respondents had direct experience with the digital functions under examination, although recruitment through digital and social networks may have resulted in a higher proportion of younger passengers.

Data were collected during November and December 2025. After the returned questionnaires had been screened, 348 valid responses were retained for analysis.

In addition to demographic questions, the questionnaire contained 21 observed indicators measured on a five-point Likert scale (Appendix). The measurement items were adapted from previous studies and adjusted to the context of railway passenger transport in Vietnam. All constructs included in the model were specified as reflective. The questionnaire focused on the user-facing digital layer of the railway transport system. Digital Experience was operationalized through railway-specific digital touchpoints covering timetable, fare, and journey-status information; onboard connectivity; digital customer support; personalized journey suggestions; and real-time interaction and response. These indicators captured passengers' evaluations of the digital functions encountered across different stages of the railway journey.

The minimum sample size was determined in accordance with power-analysis recommendations for PLS-SEM [42]. At a significance level of 5% and a statistical power of 80%, a minimum of 124 observations was required to detect an R^2 value of approximately 0.10. The final sample of 348 valid responses exceeded this requirement and was considered adequate for estimating the proposed model.

PLS-SEM was used to estimate the relationships among the constructs in the proposed model. The measurement and structural models were analyzed using SmartPLS v.4.1.1.6. The measurement model was evaluated for indicator reliability, internal consistency, convergent validity, and discriminant validity. Discriminant validity was assessed using the heterotrait-monotrait ratio of correlations (HTMT). HTMT values below 0.90 were considered acceptable, and the bootstrapped 95% confidence intervals were examined to confirm that they did not include 1.00. The structural model was assessed using path coefficients, R^2 , f^2 , and variance inflation factor (VIF) values. The mediating roles of perceived value and digital trust were examined through the specific indirect effects and their bootstrapped confidence intervals.

3 Results

The final analytical dataset comprised 348 valid observations. Regarding gender, female respondents represented 64.94%, while male respondents accounted for 35.06%. Regarding age, respondents aged 18–34 formed the dominant group and represented 81.32% of the sample (Table 1). Consequently, the study population is heavily skewed towards a younger demographic. This segment is generally more familiar with online ticketing and may place greater value on digital experiences and related technological benefits.

Table 1. Descriptive statistics

Category		Frequency	Percentage (%)
Gender	Female	212	64.94
	Male	136	35.06
Age	18–34 years	283	81.32
	35–55 years	52	14.94
	55+ years	13	3.74

Indicator quality was evaluated using outer loadings. For an indicator to demonstrate adequate relevance, an outer loading of 0.70 or higher is generally recommended. The outer loadings reported in Table 2 show that most indicators exceed 0.70 and therefore demonstrate adequate reliability in representing their latent constructs. Specifically, indicators DE1–DE7 have outer loadings ranging from 0.716 to 0.831, indicating that the items adequately capture the characteristics of Digital Experience. Perceived value, measured by four indicators, demonstrates strong convergence with high outer loadings (ranging from 0.746 to 0.864). The outer loadings of RPS1–RPS5 range from 0.768 to 0.841, suggesting a high degree of convergence among the five indicators representing RPS. For digital trust, indicator DT3 has an outer loading of 0.677, which is slightly below the 0.70 threshold. Despite this lower loading, the Digital

Trust construct still demonstrates satisfactory composite reliability ($\rho_c = 0.861$) and convergent validity (Average Variance Extracted—AVE is 0.554). Therefore, DT3 was retained to maintain the construct's content validity.

The internal consistency and convergent validity of the measurement scales were evaluated using Cronbach's alpha, Composite reliability ρ_a , and Composite reliability ρ_c (CR). According to established guidelines, values between 0.60 and 0.70 are acceptable for exploratory research, whereas the optimal range is 0.70 to 0.90 [43]. As illustrated in Table 3, all constructs exhibited robust reliability, with Cronbach's alpha coefficients ranging from 0.797 to 0.892. This indicates a strong cohesion among the observed items within each scale. Furthermore, the Composite Reliability values surpassed the recommended threshold of 0.70, with a minimum recorded value of 0.803. These findings confirm that the latent variables possess high reliability, thereby ensuring the stability and robustness of the measurement model.

Table 2. Outer loadings of the measurement model

	DE	DT	PV	RPS
DE1	0.716	–	–	–
DE2	0.809	–	–	–
DE3	0.767	–	–	–
DE4	0.748	–	–	–
DE5	0.765	–	–	–
DE6	0.811	–	–	–
DE7	0.831	–	–	–
DT1	–	0.740	–	–
DT2	–	0.823	–	–
DT3	–	0.677	–	–
DT4	–	0.757	–	–
DT5	–	0.717	–	–
PV1	–	–	0.746	–
PV2	–	–	0.817	–
PV3	–	–	0.843	–
PV4	–	–	0.864	–
RPS1	–	–	–	0.841
RPS2	–	–	–	0.790
RPS3	–	–	–	0.816
RPS4	–	–	–	0.793
RPS5	–	–	–	0.768

Note: DE—digital experience; DT—digital trust; PV—perceived value; RPS—Railway Passenger Satisfaction; an endash indicates “no data” or “not applicable”.

The AVE in Table 3 indicates the extent to which a measure is positively associated with alternative measures of the same construct. AVE values of 0.50 or higher indicate that each construct accounts for at least half of the variance in its indicators and therefore satisfies the criterion for convergent validity.

Table 3. Construct reliability and convergent validity

	Cronbach's Alpha	Composite Reliability (ρ_a)	Composite Reliability (ρ_c)	Average Variance Extracted (AVE)
DE	0.892	0.893	0.915	0.607
DT	0.797	0.803	0.861	0.554
PV	0.835	0.837	0.890	0.670
RPS	0.861	0.865	0.900	0.643

Note: DE—digital experience; DT—digital trust; PV—perceived value; RPS—Railway Passenger Satisfaction.

Discriminant validity was assessed using the HTMT criterion. The HTMT values ranged from 0.506 to 0.835, all below 0.90, and none of the bootstrapped 95% confidence intervals included 1.00. Therefore, the four constructs demonstrated satisfactory discriminant validity.

To evaluate potential multicollinearity among the indicators, the VIF was calculated. As reported in Table 4, all indicators exhibit VIF values below 5 (ranging from 1.625 to 2.648), indicating that multicollinearity is not a concern in this model. The measurement indicators therefore provide an appropriate basis for analyzing the structural relationships among constructs.

The overall measurement-model assessment confirmed that all observed indicators and latent constructs satisfactorily established reliability, convergent validity, and discriminant validity. Furthermore, the analysis verified the absence of multicollinearity. Based on these results, the structural model was then evaluated to test the hypothesized relationships specified in the research framework.

Table 4. Collinearity statistics

	VIF
DE1	1.625
DE2	2.146
DE3	1.898
DE4	1.851
DE5	2.137
DE6	2.343
DE7	2.648
DT1	1.497
DT2	1.940
DT3	1.377
DT4	1.704
DT5	1.509
PV1	1.616
PV2	2.018
PV3	2.121
PV4	2.346
RPS1	2.378
RPS2	1.993
RPS3	1.978
RPS4	1.853
RPS5	1.664

Note: VIF—Variance Inflation Factor, DE—digital experience, DT—digital trust, PV—perceived value, RPS—Railway Passenger Satisfaction.

To examine potential collinearity among the independent constructs (inner VIF), all VIF values were below 3 (Table 5), indicating that the structural model is not affected by collinearity issues. Specifically, the effects of Digital Experience on Digital Trust and Perceived Value both show VIF values of 1.000. For the paths predicting RPS from DE, DT, and PV, the highest VIF value is 2.153. These results indicate that the independent constructs provide sufficiently distinct explanatory information for the dependent construct.

Table 5. Structural model collinearity assessment

	VIF
DE → DT	1.000
DE → PV	1.000
DE → RPS	2.153
DT → RPS	1.612
PV → RPS	1.622

Note: VIF—Variance Inflation Factor, DE—digital experience, DT—digital trust, PV—perceived value, RPS—Railway Passenger Satisfaction.

The results of the structural path analysis and hypothesis testing are summarized in Table 6. The relationships among the model's constructs were evaluated based on *t*-statistics and *p*-values. A structural path is considered statistically significant at the 95% confidence level when the *t*-statistic exceeds 1.96, and the *p*-value is less than 0.05. Digital experience exerted a substantial positive influence on the mediating variables, digital trust and perceived value, with standardized path coefficients of 0.614 and 0.617, respectively. Regarding RPS, digital experience had the largest direct path coefficient ($\beta = 0.452$), followed by perceived value ($\beta = 0.283$) and digital trust ($\beta = 0.184$). All direct paths were statistically significant at $p < 0.001$.

The coefficient of determination (R^2) indicates the proportion of variance in the dependent construct explained by the independent constructs. The results reported in Table 7 indicate that the proposed model demonstrates substantial explanatory power for RPS, with an R^2 of 0.622. This implies that digital experience, digital trust, and perceived value jointly explain 62.2% of the variance in RPS. The mediating constructs digital trust and perceived value yield R^2 values of 0.377 and 0.381, respectively, suggesting that digital experience explains a substantial proportion of

variance in both mediators. In addition, the adjusted R^2 values are close to the corresponding R^2 values, indicating little reduction after adjustment for model complexity.

Table 6. Structural path coefficients and hypothesis testing

	Original Sample (O)	Sample Mean (M)	Standard Deviation (STDEV)	<i>t</i> -Statistics	<i>p</i> -Values
DE → DT	0.614	0.615	0.046	13.440	0.000
DE → PV	0.617	0.619	0.038	16.417	0.000
DE → RPS	0.452	0.450	0.060	7.579	0.000
DT → RPS	0.184	0.186	0.049	3.730	0.000
PV → RPS	0.283	0.283	0.053	5.396	0.000

Note: DE—digital experience; DT—digital trust; PV—perceived value; RPS—Railway Passenger Satisfaction.

Table 7. R^2 and adjusted R^2 values of the endogenous constructs

	R^2	R^2 Adjusted
DT	0.377	0.376
PV	0.381	0.379
RPS	0.622	0.619

Note: DE—digital experience, DT—digital trust, PV—perceived value, RPS—Railway Passenger Satisfaction.

The effect size (f^2) assesses the practical magnitude of each exogenous construct's contribution to the variance explained in the endogenous construct. Table 8 reports the relative role of each construct in the structural model. The effects of digital experience on the mediators are substantial, with $f^2 = 0.606$ for digital trust and $f^2 = 0.616$ for perceived value. For RPS, digital experience has a medium effect ($f^2 = 0.251$), perceived value has a small effect approaching the medium threshold ($f^2 = 0.131$), and digital trust has a small effect ($f^2 = 0.055$). These results indicate that the direct contribution of digital experience to satisfaction is comparatively stronger, while both mediators provide additional explanatory value.

Table 8. f^2 matrix

	DE	DT	PV	RPS
DE	—	0.606	0.616	0.251
DT	—	—	—	0.055
PV	—	—	—	0.131
RPS	—	—	—	—

Note: DE—digital experience, DT—digital trust, PV—perceived value, RPS—Railway Passenger Satisfaction.

The mediation analysis was conducted to isolate and evaluate the indirect pathways operating through the two intervening variables: digital trust and perceived value. Table 9 delineates the results of the specific indirect effects assessment. Specifically, the pathway mediated by perceived value exhibited a specific indirect effect coefficient (β) of 0.175. Concurrently, the pathway mediated by digital trust demonstrated an indirect effect of 0.113. Crucially, both mediating mechanisms were confirmed to be highly statistically significant ($p < 0.001$).

Table 9. Evaluating the specific indirect effects

	Original Sample (O)	Sample Mean (M)	Standard Deviation (STDEV)	<i>t</i> -Statistics	<i>p</i> -Values
DE → PV → RPS	0.175	0.175	0.034	5.198	0.000
DE → DT → RPS	0.113	0.115	0.034	3.368	0.001

Note: DE—digital experience; DT—digital trust; PV—perceived value; RPS—Railway Passenger Satisfaction.

The assessment of total indirect effects indicates the combined mediating influence of perceived value and digital trust in transmitting the effect of the independent construct digital experience to the dependent construct RPS. As

reported in Table 10, the total indirect effect of digital experience on RPS is 0.288 and is highly statistically significant ($p = 0.000$), providing strong evidence for the presence of mediation in the proposed model.

Table 10. Total indirect effects

	Original Sample (O)	Sample Mean (M)	Standard Deviation (STDEV)	<i>t</i> -Statistics	<i>p</i> -Values
DE → RPS	0.288	0.290	0.046	6.274	0.000

Note: DE—digital experience; RPS—Railway Passenger Satisfaction.

4 Discussion

The measurement and structural model results showed that the proposed model explained 62.2% of the variance in RPS. Digital experience was closely associated with passengers' evaluations of both perceived value and digital trust, which in turn were related to their satisfaction with railway services. Digital experience also explained 38.1% and 37.7% of the variance in perceived value and digital trust, respectively. These results show that passengers' assessment of the user-facing digital layer of the railway transport system involves more than the availability of individual functions. It also reflects whether interactions with ticketing, payment, information, connectivity, and support services are considered worthwhile and dependable.

Digital experience had a positive and statistically significant association with RPS ($\beta = 0.452$, $p < 0.001$), with a medium effect size ($f^2 = 0.251$). This result is consistent with previous evidence that railway applications, online ticketing, real-time passenger information, and electronic service quality are related to passenger convenience and satisfaction [16, 17, 22, 33]. From a passenger-centered system perspective, the result indicates that digital functions are not experienced as separate technical additions to the core transport service. Instead, they form part of the operational interface through which passengers plan journeys, complete transactions, obtain information, and respond to changes in railway operations. The result is also consistent with electronic service quality theory, which identifies efficiency and fulfillment as important elements of users' overall service quality assessments [15].

In the Vietnamese railway context, the positive association may reflect the contrast between conventional service processes and the expanding digital service environment. Online booking, mobile payment, electronic tickets, and real-time information reduce the need for physical transactions and give passengers more direct access to journey-related information. However, the demographic composition of the sample should be considered when interpreting the strength of this relationship. Respondents aged 18–34 represented more than four-fifths of the sample. The estimated association may therefore reflect, in part, the evaluations of younger passengers who are more accustomed to online booking, mobile payment, and real-time information services. The result should not automatically be generalized to older passenger groups or passengers with limited experience of digital technology.

The mediation analysis showed that perceived value significantly mediated the relationship between digital experience and RPS, with an indirect effect of $\beta = 0.175$ ($p < 0.001$). This pathway was stronger than the corresponding pathway through digital trust, for which the indirect effect was $\beta = 0.113$ ($p = 0.001$). The result is consistent with previous railway studies identifying perceived value as an important factor in passenger satisfaction and loyalty [34–36]. It suggests that part of the relationship between digital experience and RPS operates through passengers' assessment of what they receive from the digital railway service in relation to the time, effort, cost, and uncertainty involved in using it.

The direct path from digital experience to RPS remained positive and statistically significant after perceived value was included in the model, indicating partial complementary mediation. In the Vietnamese context, this result may be associated with the practical changes produced by the transition from paper tickets and station-based transactions to electronic tickets, online payment, and mobile access to journey information. These functions can reduce transaction time, simplify travel preparation, and give passengers greater control over their journeys. The stronger indirect pathway through perceived value indicates that convenience, time savings, and value for money play a particularly important role in passengers' evaluations of digital railway services. For the implementation of digital railway systems, this finding means that adding further functions is unlikely to be sufficient unless passengers can recognize a clear practical value in using them.

Digital trust also had a significant mediating role, with a specific indirect effect of 0.113 ($p = 0.001$). Although this indirect effect was smaller than that of perceived value, digital trust remained relevant to passenger satisfaction in the digital service environment. Digital experience was strongly associated with digital trust ($\beta = 0.614$, $p < 0.001$), while digital trust was positively associated with RPS ($\beta = 0.184$, $p < 0.001$). These relationships are particularly relevant to digital railway services because passengers must rely on the system to process payments accurately, protect personal information, issue valid electronic tickets, and provide consistent operational information.

Passengers are more likely to evaluate digital railway services positively when transactions are secure, personal information is handled appropriately, and the services delivered correspond to the information displayed through digital channels. Conversely, inconsistent information, failed payments, unstable applications, or unclear privacy practices may weaken trust even when the required digital functions are technically available. The present results are consistent with previous evidence reported by Dewi et al. [37], Flores et al. [38], and Wonglakorn et al. [36]. Taken together, the two mediators represent complementary dimensions of passenger-system interaction. Perceived value indicates whether passengers consider digital railway services worthwhile, whereas digital trust indicates whether they consider those services and the systems supporting them reliable and safe. Their simultaneous inclusion provides a more complete passenger-side evaluation of digital railway system implementation than either mechanism would provide separately.

5 Conclusions

This study investigated the relationship between digital experience and RPS and examined the parallel mediating roles of perceived value and digital trust. PLS-SEM analysis of survey responses from 348 passengers showed that digital experience was directly associated with RPS and was also indirectly associated with satisfaction through both mediators. The pathway through perceived value was stronger than the pathway through digital trust. These findings indicate that the passenger-side performance of digital railway transport systems depends not only on the availability of digital functions but also on whether passengers find their interactions with those functions worthwhile, reliable, and secure.

The study contributes to research on technology acceptance and electronic service quality by conceptualizing digital experience as an integrated evaluation of the user-facing digital touchpoints encountered throughout a railway journey. Rather than treating online ticketing, payment, passenger information, onboard connectivity, and customer support as unrelated services, the proposed model considers them components of the same passenger-system interaction. The simultaneous examination of perceived value and digital trust identifies two distinct mechanisms connecting this interaction with RPS. Their different mediation strengths further show that the practical value passengers obtain from digital services and their willingness to rely on those services do not play identical roles in satisfaction formation.

For railway operators, the findings indicate that digital transformation should be managed as an integrated part of the passenger transport system rather than as a series of independent technology projects. Usability, transaction efficiency, transparent fare and itinerary information, and appropriately personalized services are central to making the value of digital functions visible to passengers. At the same time, payment stability, consistency of information across channels, protection of personal data, and timely customer support are necessary for maintaining digital trust. Decisions concerning digital investment should therefore consider how each proposed function contributes to the continuity of the passenger journey, reduces the time and effort required from users, and supports confidence in the wider railway service.

This study has several limitations. First, purposive and snowball sampling may have introduced selection, self-selection, and network-related biases. Respondents aged 18–34 accounted for 81.32% of the sample, limiting its representativeness. The findings therefore largely reflect the assessments of younger passengers, who are generally more familiar with digital technology. Second, the cross-sectional design does not establish temporal or causal relationships among the constructs. The use of self-reported data may also have introduced response bias. Future studies should use probability, stratified, or quota sampling; recruit samples with a more balanced demographic composition; and employ longitudinal or multi-source data to examine whether the proposed relationships remain stable across passenger groups and over time.

Author Contributions

Conceptualization, T.T.H.; methodology, T.T.H.; validation, T.T.H. and T.N.; formal analysis, T.T.H.; investigation, T.T.H. and T.N.; data curation, T.T.H. and T.N.; writing—original draft preparation, T.T.H.; writing—review and editing, T.T.H. and T.N.; visualization, T.T.H.; supervision, T.T.H.; project administration, T.T.H. All authors have read and agreed to the published version of the manuscript.

Informed Consent Statement

Informed consent was obtained from all subjects involved in the study.

Ethical Approval

This study was conducted in accordance with the scientific research ethics regulations of the University of Transport Technology, Vietnam. Ethical review and approval were waived because the survey was anonymous and voluntary, involved adult participants, and did not collect sensitive personal data.

Data Availability

The data supporting the findings of this study are not publicly available due to privacy and ethical restrictions but may be made available from the corresponding author upon reasonable request.

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Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

ChatGPT, developed by OpenAI, and Grammarly assist in editing and refining the manuscript revision process. These tools were not used to generate research data, results, or references, nor does it replace the authors' intellectual contributions. All suggestions are carefully reviewed, verified, and edited by the authors. The authors are responsible for the accuracy, originality, integrity, and final content of the manuscript, as well as ensuring compliance with ethical and publishing standards.

References

- [1] I. Anwer, M. A. Javid, M. I. Yousuf, M. Farooq, N. Ali, S. Suparp, and Q. Hussain, "Travelers' propensity to use intercity railway services in emerging economies: Significance of passengers' satisfaction and communication technologies," *Sustainability*, vol. 16, no. 20, p. 8921, 2024. <https://doi.org/10.3390/su16208921>
- [2] F. D. Davis, "Perceived usefulness, perceived ease of use, and user acceptance of information technology," *MIS Q.*, vol. 13, no. 3, pp. 319–340, 1989. <https://doi.org/10.2307/249008>
- [3] F. D. Davis, "User acceptance of information technology: System characteristics, user perceptions and behavioral impacts," *Int. J. Man-Machine Stud.*, vol. 38, no. 3, pp. 475–487, 1993. <https://doi.org/10.1006/imms.1993.1022>
- [4] V. Venkatesh and F. D. Davis, "A theoretical extension of the technology acceptance model: Four longitudinal field studies," *Manag. Sci.*, vol. 46, no. 2, pp. 186–204, 2000. <https://doi.org/10.1287/mnsc.46.2.186.11926>
- [5] D. Marikyan and S. Papagiannidis, *Technology Acceptance Model: A Review*. TheoryHub Book, 2023.
- [6] V. Venkatesh and H. Bala, "Technology acceptance model 3 and a research agenda on interventions," *Decis. Sci.*, vol. 39, no. 2, pp. 273–315, 2008. <https://doi.org/10.1111/j.1540-5915.2008.00192.x>
- [7] V. A. Zeithaml, "Consumer perceptions of price, quality, and value: A means-end model and synthesis of evidence," *J. Mark.*, vol. 52, no. 3, pp. 2–22, 1988. <https://doi.org/10.1177/002224298805200302>
- [8] A. Eggert and W. Ulaga, "Customer perceived value: A substitute for satisfaction in business markets?" *J. Bus. Ind. Mark.*, vol. 17, no. 2–3, pp. 107–118, 2002. <https://doi.org/10.1108/08858620210419754>
- [9] J. C. Sweeney and G. N. Soutar, "Consumer perceived value: The development of a multiple item scale," *J. Retail.*, vol. 77, no. 2, pp. 203–220, 2001. [https://doi.org/10.1016/S0022-4359\(01\)00041-0](https://doi.org/10.1016/S0022-4359(01)00041-0)
- [10] D. H. McKnight, V. Choudhury, and C. Kacmar, "Developing and validating trust measures for e-commerce: An integrative typology," *Inf. Syst. Res.*, vol. 13, no. 3, pp. 334–359, 2002. <https://doi.org/10.1287/isre.13.3.334.81>
- [11] K. Siau and Z. Shen, "Building customer trust in mobile commerce," *Commun. ACM*, vol. 46, no. 4, pp. 91–94, 2003. <https://doi.org/10.1145/641205.641211>
- [12] R. Ladhari, "Service quality, emotional satisfaction, and behavioural intentions: A study in the hotel industry," *Manag. Serv. Qual. Int. J.*, vol. 19, no. 3, pp. 308–331, 2009. <https://doi.org/10.1108/09604520910955320>
- [13] A. Parasuraman, V. A. Zeithaml, and L. L. Berry, "SERVQUAL: A multiple-item scale for measuring consumer perceptions of service quality," *J. Retailing*, vol. 64, no. 1, pp. 12–37, 1988.
- [14] W. H. DeLone and E. R. McLean, "The DeLone and McLean model of information systems success: A ten-year update," *J. Manag. Inf. Syst.*, vol. 19, no. 4, pp. 9–30, 2003. <https://doi.org/10.1080/07421222.2003.11045748>
- [15] A. Parasuraman, V. A. Zeithaml, and A. Malhotra, "ES-QUAL: A multiple-item scale for assessing electronic service quality," *J. Serv. Res.*, vol. 7, no. 3, pp. 213–233, 2005. <https://doi.org/10.1177/1094670504271156>
- [16] O. S. Waruwu, P. Andini, S. S. Waruwu, F. Zebua, D. C. D. Crisanta, and D. F. F. D. B. Butar, "The impact of system quality, information quality, and service quality of rail ticketing system (RTS) on customer satisfaction of PT. Kereta Api Indonesia," *Glob. Insights Manag. Econ. Res.*, vol. 1, no. 2, pp. 33–37, 2025. <https://doi.org/10.53905/Gimer.v1i02.6>
- [17] J. V. Stanly, J. Sheeba, and K. A. Perumal, "An empirical study on the impact of the Rail One App on passenger convenience and service experience: A SEM approach," *Adv. Consum. Res.*, vol. 2, no. 4, pp. 2931–2939, 2025.

- [18] Z. Kljaić, D. Pavković, M. Cipek, M. Trstenjak, T. J. Mlinarić, and M. Nikšić, "An overview of current challenges and emerging technologies to facilitate increased energy efficiency, safety, and sustainability of railway transport," *Futur. Internet*, vol. 15, no. 11, p. 347, 2023. <https://doi.org/10.3390/fi15110347>
- [19] A. Ø. Sørensen, N. O. Olsson, M. M. Akhtar, and H. Bull-Berg, "Approaches, technologies and importance of analysis of the number of train travellers," *Urban Plan. Transp. Res.*, vol. 7, no. 1, pp. 1–18, 2019. <https://doi.org/10.1080/21650020.2019.1566022>
- [20] E. Mogaji and N. P. Nguyen, "Evaluating the emergence of contactless digital payment technology for transportation," *Technol. Forecast. Soc. Chang.*, vol. 203, p. 123378, 2024. <https://doi.org/10.1016/j.techfore.2024.123378>
- [21] Z. Cao, T. Xu, S. Zhang, A. A. Ceder, and Y. Sun, "Comparative evaluation: Passenger satisfaction and operation efficiency of different even-headway and even-load public transport timetables," *KSCE J. Civ. Eng.*, vol. 26, no. 7, pp. 3063–3081, 2022. <https://doi.org/10.1007/s12205-022-1231-x>
- [22] B. Henriquez-Jara, J. Arriagada, and A. Tirachini, "Impact of real-time information on passenger satisfaction across varying public transport quality levels in 13 Chilean cities," *Transp. Res. Part A: Policy Pr.*, vol. 200, p. 104622, 2025. <https://doi.org/10.1016/j.tra.2025.104622>
- [23] X. Xu, J. Cong, and T. Xia, "A study of the impact of the use of real-time crowding information on the perceived service quality of high-speed rail by passengers with unfixed-seat tickets," *Appl. Sci.*, vol. 14, no. 2, p. 535, 2024. <https://doi.org/10.3390/app14020535>
- [24] Y. Zhang, E. Jenelius, and K. Kottenhoff, "Impact of real-time crowding information: A Stockholm metro pilot study," *Public Transp.*, vol. 9, no. 3, pp. 483–499, 2017. <https://doi.org/10.1007/s12469-016-0150-y>
- [25] W. Chow, D. Block-Schachter, and S. Hickey, "Impacts of real-time passenger information signs in rail stations at the Massachusetts Bay Transportation Authority," *Transp. Res. Rec. J. Transp. Res. Board*, vol. 2419, no. 1, pp. 1–10, 2014. <https://doi.org/10.3141/2419-01>
- [26] K. E. Watkins, B. Ferris, A. Borning, G. S. Rutherford, and D. Layton, "Where is my bus? Impact of mobile real-time information on the perceived and actual wait time of transit riders," *Transp. Res. Part A: Policy Pr.*, vol. 45, no. 8, pp. 839–848, 2011. <https://doi.org/10.1016/j.tra.2011.06.010>
- [27] E. Malichová, Y. Cornet, and M. Hudák, "Travellers' use and perception of travel time in long-distance trips in Europe," *Travel Behav. Soc.*, vol. 27, pp. 95–106, 2022. <https://doi.org/10.1016/j.tbs.2021.12.003>
- [28] R. Chen, A. Shalaby, and D. Da Silva, "Trends in Toronto's subway ridership recovery: An exploratory analysis of Wi-Fi records," *Transp. Res. Rec. J. Transp. Res. Board*, vol. 2678, no. 11, pp. 1156–1176, 2024. <https://doi.org/10.1177/03611981241242775>
- [29] I. Grgurević, K. Juršić, and V. Rajič, "Overview of Wi-Fi-based automatic passenger counting solutions in public urban transport," in *Sustainable Management of Manufacturing Systems in Industry 4.0, EAI/Springer Innovations in Communication and Computing*. Cham: Springer, 2022, pp. 181–196. https://doi.org/10.1007/978-3-030-90462-3_12
- [30] Z. Zheng, S. Washington, P. Hyland, K. Sloan, and Y. Liu, "Preference heterogeneity in mode choice based on a nationwide survey with a focus on urban rail," *Transp. Res. Part A: Policy Pr.*, vol. 91, pp. 178–194, 2016. <https://doi.org/10.1016/j.tra.2016.06.032>
- [31] C. Calastri, J. Pawlak, and R. Batley, "Participation in online activities while travelling: An application of the MDCEV model in the context of rail travel," *Transportation*, vol. 49, no. 1, pp. 61–87, 2022. <https://doi.org/10.1007/s11116-021-10166-8>
- [32] S. Naumann and F. Wolf, "Personalised passenger notification in the case of incidents," in *International Conference on Transport Systems Telematics, Communications in Computer and Information Science (vol. 239)*. Berlin, Heidelberg: Springer, 2011, pp. 306–313. https://doi.org/10.1007/978-3-642-24660-9_35
- [33] R. A. Triani and F. S. Schouten, "Analysis of the application of online ticket booking application access by KAI," *J. Bus. Stud. Manage. Rev.*, vol. 7, no. 1, pp. 132–138, 2023. <https://doi.org/10.22437/jbsmr.v7i1.29178>
- [34] X. C. Vuong, T. M. T. Vu, D. A. Trieu, and T. T. Vu, "Factors influencing customer satisfaction and loyalty towards urban railway services in Hanoi: Mediating role of perceived value," *PLoS ONE*, vol. 21, no. 1, p. e0340760, 2026. <https://doi.org/10.1371/journal.pone.0340760>
- [35] R. Lieophairot, N. Rojniruttikul, and S. Chaveesuk, "Factors influencing rail service passenger loyalty among older Thai adults," *Sustainability*, vol. 17, no. 18, p. 8240, 2025. <https://doi.org/10.3390/su17188240>
- [36] N. Wonglakorn, V. Ratanavaraha, A. Karoonsoontawong, and S. Jomnonkwao, "Exploring passenger loyalty and related factors for urban railways in Thailand," *Sustainability*, vol. 13, no. 10, p. 5517, 2021. <https://doi.org/10.3390/su13105517>
- [37] A. S. Dewi, A. Wulandari, A. Rahayu, and H. Hendrayati, "Gaining user satisfaction of KAI-Access: E-service quality dimensions as antecedent through e-trust," *Asia Pac. Manag. Bus. Appl.*, vol. 11, no. 3, pp. 363–378, 2023. <https://doi.org/10.21776/ub.apmba.2023.011.03.8>

- [38] L. C. Flores, A. K. S. Ong, R. A. G. Roque IV, T. M. C. Palad, J. D. D. Concepcion, and R. D. Aguas Jr., "Assessment of service quality and trust of e-public transportation in Doha Qatar," *World Electr. Veh. J.*, vol. 16, no. 3, p. 174, 2025. <https://doi.org/10.3390/wevj16030174>
- [39] I. Etikan, S. A. Musa, and R. S. Alkassim, "Comparison of convenience sampling and purposive sampling," *Am. J. Theor. Appl. Stat.*, vol. 5, no. 1, pp. 1–4, 2016. <https://doi.org/10.11648/j.ajtas.20160501.11>
- [40] K. Browne, "Snowball sampling: Using social networks to research non-heterosexual women," *Int. J. Soc. Res. Methodol.*, vol. 8, no. 1, pp. 47–60, 2005. <https://doi.org/10.1080/1364557032000081663>
- [41] J. Simkus, "Snowball Sampling Method: Techniques & Examples," 2023. <https://www.simplypsychology.org/snowball-sampling.html>
- [42] J. F. Hair, G. T. M. Hult, C. M. Ringle, M. Sarstedt, N. P. Danks, and S. Ray, "Partial least squares structural equation modeling (PLS-SEM) using R: A workbook," in *Classroom Companion: Business*. Cham: Springer International Publishing, 2021. <https://doi.org/10.1007/978-3-030-80519-7>
- [43] J. C. Nunnally and I. H. Bernstein, *Psychometric Theory (3rd ed.)*. New York, NY, USA: McGraw-Hill, 1994.

Appendix

Table A1. Survey items

Symbol	Scales	Sources
Digital experience (DE)		
DE1	The app/website is easy to navigate and allows me to complete transactions quickly.	[6], Authors' development
DE2	Information regarding train schedules, fares, and trip status is accurate and timely.	[14, 22, 25], Authors' development
DE3	The system is stable, with reliable in-train Wi-Fi and real-time connectivity during the journey.	[14, 16, 28], Authors' development
DE4	Digital customer support effectively handles my inquiries and technical issues.	[14, 15, 37], Authors' development
DE5	The platform provides personalized travel suggestions and offers based on my history.	[32, 37], Authors' development
DE6	I can easily interact with the service provider through two-way channels (chatbots, feedback).	[15, 37, 38], Authors' development
DE7	The system provides immediate responses and real-time updates for my requests.	[15, 22, 26], Authors' development
Perceived value (PV)		
PV1	Using digital services is convenient, saves time, and helps me achieve my travel goals.	[9], Authors' development
PV2	I feel secure and comfortable when using the company's digital services.	[9, 12], Authors' development
PV3	Using modern digital services enhances my social status/image as a savvy traveler.	[9], Authors' development
PV4	The service provides good value for money through discounts and overall cost savings.	[9, 34], Authors' development
Digital trust (DT)		
DT1	I trust that the digital platform processes my transactions accurately.	[10, 37], Authors' development
DT2	I believe the railway company is honest regarding its information and policies.	[10, 37], Authors' development
DT3	I trust that the platform genuinely cares about the best interests of its customers.	[10, 37], Authors' development
DT4	I am confident that my personal information is kept private and confidential.	[37, 38], Authors' development
DT5	I believe the online payment system is secure and safe to use.	[11, 20], Authors' development
Railway Passenger Satisfaction (RPS)		
RPS1	I am satisfied with my digital booking and payment experience.	[16, 17, 37], Authors' development
RPS2	I am satisfied with the trip-related information provided by the company.	[22, 25, 26], Authors' development
RPS3	I am satisfied with the customer support services I received.	[15, 37], Authors' development
RPS4	I am satisfied with the railway company's commitment to system safety.	[1, 34, 36], Authors' development
RPS5	Overall, I am completely satisfied with the railway service.	[1, 35, 36], Authors' development